RESOLUTION MSC.32(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING LIQUEFIED GASES IN BULK (IGC CODE) RESOLUTION MSC.32(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING LIQUEFIED GASES IN BULK (IGC CODE)
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the fUnctions of the Committee,
RECALLING ALSO resolution MSC.5(48), by which the Committee adopted the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code),
RECALLING FURTHER article VIII(b) and regulation VII/ll.1 of the International Convention for the Safety of Life at Sea (SaLAS), 1974, as amended, concerning the procedure for amending the IGC Code,
BEING DESIROUS of keeping the IGC Code up to date.
HAVING CONSIDERED, at its sixty - third session, amendments to the Code proposed and circulated in accordance with article VIII(b)(i) of the SaLAS Convention,

BEING DESIROUS of keeping the IGC Code up to date.
HAVING CONSIDERED, at its sixty - third session, amendments to the Code proposed and circulated in accordance with article VIII(b)(i) of the SaLAS Convention,
1. ADOPTS, in accordance with article VIII(b)(iv) of the SaLAS Convention. amendments to the Code, the text of which is set out in the Annex to the present resolution;
2. DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the SOLAS Convention, that the amendments shall be deemed to have been accepted on 1 January 1998 unless, prior to that date, more than one third of the Contracting Governments to the SaLAS Convention, or Contracting Governments the combined merchant fleets of which constitute not less than fifty per cent of the gross tonnage of the world's merchant fleet, have notified their Objections to the amendments;
3. INVITES Contracting Governments to note that, in accordance with article VIII(b)(vii)(2) of the SaLAS Convention, the amendments shall enter into force on 1 July 1998 upon their acceptance in accordance with paragraph 2 above;
4. REQUESTS the Secretary-General, in conformity with article VIII(b)(v) of

article VIII(b)(vii)(2) of the SaLAS Convention, the amendments shall enter into force on 1 July 1998 upon their acceptance in accordance with paragraph 2 above;

4. REQUESTS the Secretary-General, in conformity with article VIII(b)(v) of the SaLAS Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the Annex to all Contracting Governments to the SaLAS Convention;

5. FURTHER REQUESTS the Secretary-General to transmit copies of the resolution and its Annex to Members of the Organization which are not Contracting Governments to the SOLAS Convention.

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ANNEX

AMENDMENTS TO THE INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING LIQUEFIED GASES IN BULK (IGC CODE)

Amendments related to application

1 Existing paragraphs 1.1.2 and 1.1.3 are repla ce d by the following:

"1.1. 2 Unless expressly provided otherwise, the Code applies. to s hips the keels of which are laid or which are at a stage at whi ch:

Amendments related to application

1 Existing paragraphs 1.1.2 and 1.1.3 are repla ce d by the following:

"1.1. 2 Unless expressly provided otherwise, the Code applies. to s hips the keels of which are laid or which are at a stage at whi ch:

.1 construction identifiable with the ship begins; and

.2 assembly of that ship has commenced comprising at least 50 teunes or l' of the estimated mass of all structural mate rial, whichever is the less;

on or after 1 July 1998. Ships constructed before 1 July 1998 are to comply with resolution MSC.5(48) adopted on 1 7 June 1963 subject to amendment s by r e solution MSC.30(61) adopted on 11 December 1992.

1.1.3 A ship. irrespective of the date of construction, which is converted to a gas carrier on or after 1 July 1998, should be treated as a gas carrier constructed on the date on which such conversion commences."

Amendments related to fillipg limits

2 The existing chapter 15 is replaced by the following:

"CHAPTER 15

FILLING LIMITS FOR CARGO TANKS

15.1 General

15.1.1 No cargo tanks should have a higher filling limit (FL) than 98' at the reference temperature, except as permitted by 15.1. 3.

"CHAPTER 15
FILLING LIMITS FOR CARGO TANKS
15.1 General
15.1.1 No cargo tanks should have a higher filling limit (FL) than 98' at the reference temperature, except as permitted by 15.1. 3.
15.1.2 The maximum loading limit (LL) to which a cargo tank may be loaded should be determined by the following formula:
LL
where:
LL
FL
PR
FL
PL
loading limit exp r essed as a pe r centage, being the maximum allowable liguid volume rela tive to the tank volume to which the tank may be loaded;
filling limit as specified i n 15.1.1 or 15.1.3;
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PL
~ 3 ~
relative density of cargo at the reference temperature: and
relative density of cargo at the loading temperature and pressure.
15.1.3 The Administration may allow a higher f illing limit (FL) than the limit of 98' specified in 15.1.1 at the reference temperature, taking int o account the shape of the tank, arrangements of pressure relief valves, accuracy of level and temperature gauging and the difference between the

limit of 98' specified in 15.1.1 at the reference temperature, taking int o account the shape of the tank, arrangements of pressure relief valves, accuracy of level and temperature gauging and the difference between the loading temperature and the temperature corresponding to the vapour pressure of the cargo at the set pressure of the pressure relief valves, provided the conditions specified in 8.2.17 are maintained.

15.1.4 For the purposes of this chapter only, "reference temperature" means:

.1 the temperature corresponding to the vapour pressure of the cargo at the set pressure of the pressure relief valves when DO cargo vapour pressure/temperature control as referred to in chapter 7 is provided;

.2 the temperature of the cargo upon termination of loading, during transport. or at unloading, whichever is the greatest. when a cargo vapour pressure/temperature control as referred to in chapter 7 is provided. If this reference temperature would result in the cargo tank becoming liquid full before the cargo reaches a temperature corresponding to the vapour pressure of the cargo at the set pressure of the relief valves required in 8.2. an additional

tank becoming liquid full before the cargo reaches a temperature
corresponding to the vapour pressure of the cargo at the set
pressure of the relief valves required in 8.2. an additional
pressure relieving system complying with 8.3 should be fitted.
15.1.5 The Administration may allow type C tanks to be loaded acco r ding
to the following formula, provided that the tank vent system has been appr oved
in accordance with 8.2.18:
LL
where:
LL
FL
PL
PR
Fe
PL
loading limit as specified in 15.1.2;
filling limit as specified in 15.1.1 or 15.1.3;
relative density of cargo at the highest temperature
which the cargo may reach upon termination of loading,
during transport, or at unloading. under the ambient
design temperature conditions described in 7.1.2; and
as specif i ed in 15.1.2.
This paragraph does not apply to products requiring a type IG ship.
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THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
LIQUEFIED GASES IN BULK (IGC CODE) - 4 -
15.2 Information to be provided to the master
The maximum a l lowable loading limits for each cargo tank should be

15.2 Information to be provided to the master

The maximum a l lowable loading limits for each cargo tank should be indicat~d for each product which may be carried, for each load ing temperature which may be applied and for the applicable maximum reference tem per a t ure, on a list to be approved by the Administration. Pressures at which the pressure relief valves, including those valves required by 8.3, have been set should also be stated on the list. A copy of the li st should be kep t permanently on board by the master.

15.3 Chapter 15 app lie s to all ships regardless of the date of const ruction. "

The fol l owing words are added at the end o f ex i sti ng pa ragraph 8 . 2.17: "at the ma:z:imum allowable filling limit (FL)",

4 The following new paragraph 8.2.18 i s added after exis ting paragraph 8.2.17:

"8.2.18 The adeguacy of the vent system fitted on tank s l oaded in accordance with 15.1 . 5 is to be demonstrated using the guidelines developed by the Organization. A relevant certificate should be

paragraph 8.2.17:
"8.2.18 The adeguacy of the vent system fitted on tank s l oaded in accordance with 15.1 . 5 is to be demonstrated using the guidelines developed by the Organization. A relevant certificate should be permanently kept on board the ship. For the purposes of this pa ragraph. vent system means:
.1 the tank outlet and the piping to the pressure relief valve;
.2 the pressure relief valve;
.3 the piping fro m the pressure relief valve to the location of discharge to the atmosphere and including any interconnections and piping which joins other tank s .
This paragraph may apply to all ships rega rdl es s of t he date of construction. "
Amendments r e lated to cargg tank vent system s
5 The existing paragraph 8.2.3 is replaced by the foll owin g :
"8.2.3 In general, the setting of the pressure relief valves should not be higher than the vapour pressure which ha s been used in t he design of the tank. However. where two or more pressure relief valves are fitted, va l ves comprising not more than 5o, of the total relieving capa city may be set at a pressure up to 5'\, above HARVS."
6 The following sentences are added to existing paragraph 8.2.4:

va l ves comprising not more than 5o, of the total relieving capa city may be set at a pressure up to 5'\, above HARVS."

6 The following sentences are added to existing paragraph 8.2.4:

"Valves should be constructed of materia ls with a melti ng point above 925°C. Consideration should be given to lower melting point materials for internal parts and seal s if their use wi ll yield a significant improvement in the general operation of the valve."

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7 The existing paragraph 8.2.9 i s replaced by the following:

"8.2.9 Each pressure relief valve installed on a cargo tank should be connected to a venting system, which should be 50 constructed that the discharge of gas will be unimpeded and directed vertically upwards at the exit, and so arranged as to minimize the possibility of water or snow entering the vent system. The height of vent exits should be not less than B/3 or 6 m, whichever is the greater, above the weather deck and 6 m

exit, and so arranged as to minimize the possibility of water or snow entering the vent system. The height of vent exits should be not less than B/3 or 6 m, whichever is the greater, above the weather deck and 6 m above the working area, the fore and aft gangway, deck storage tanks and cargo liquid lines."

8 The following sentences are added to existing paragraph 8.2.16:

"The pressure drop in the vent line from the tank to the pressure relief valve inlet should not exceed 3~ of the valve set pressure. For unbalanced pressure relief valves the back pressure in the discharge line should not exceed 10~ of the gauge pressure at the relief valve inlet with the vent lines under fire exposure as referred to in 8.5.2."

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